

Notes on Markov Decision Processes

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Question: Why do sigmoid and binary cross-entropy go together?

Answer:

Sigmoid activation and binary cross-entropy loss are commonly used together for binary classification problems because there are only two classes, usually represented as 0 and 1.

1. Sigmoid Activation Function

The sigmoid function converts any real-valued number into a probability between 0 and 1:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

This makes it useful for predicting the probability of belonging to class 1.

Examples:

0.95 → Very likely class 1

0.02 → Very likely class 0

0.50 → Uncertain

2. Binary Cross-Entropy Loss

Binary cross-entropy measures how different the predicted probability is from the true label.

The loss function is:

$$L = -(y \log(p) + (1 - y) \log(1 - p))$$

Where:

- y = true label (0 or 1)
- p = predicted probability from the sigmoid function

3. Why They Work Together

Sigmoid outputs probabilities in the range:

$$0 \leq p \leq 1$$

Binary cross-entropy is specifically designed to evaluate probability predictions for two classes.

Therefore, sigmoid naturally provides the type of output that binary cross-entropy expects.

4. Typical Keras Example

```
model.add(Dense(1, activation='sigmoid'))
```

```
model.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)
```

This setup means:

- One output neuron
- Output is a probability
- Classification between classes 0 and 1
- Training uses binary cross-entropy loss